Tilt Blink, 47
schematic circuit, 48–49
source code summary, 50–52
uploading source code, 49–50
Declaring functions, 109–110
Delay() function, 8, 21, 68, 86, 96–97
DelayMicroseconds() function, 97
DetachInterrupt() function, 118
Digital functions
digitalRead(), 70
digitalWrite(), 69–70
pinMode(), 68–69
Digital Ins and Outs
analog to digital conversion (ADC), 61
Arduino I/O
current limiting resistor, 63
floating pin, 64
polarized components, 63
sink, 62
sinking current, 63
source, 62
sourcing current, 63
transistors, 63
digital functions
digitalRead(), 70
digitalWrite(), 69–70
pinMode(), 68–69
Noisy Cricket
schematic circuit, 64–65
source code summary, 66–68
uploading source code, 65–66
pulse width modulation (PWM), 61
state changes
counting, 73–74
edge detection, 71
modality, 75–76
pushbutton circuit, 71–72
toggle, 72–73
DigitalRead() function, 70
DigitalWrite() function, 8, 21, 25, 69–70, 206
Distributed hardware development model, 5
Do statement, 57–58
Double servo sweep, 160
Driver installation on Windows 7, 14
Duty cycle, 80–81
Downloading and installing Arduino software, 12–13

E

Electronically erasable program read-only memory (EEPROM), 137–138
Electronics
capacitors, 225–226
circuits
components, 222
conductors, 222
insulators, 223
power source, 222
schematic circuit, 222, 231
schematic symbols, 232
components, 242
diodes, 226–227
light-emitting diode, 227
rectifier diode, 227
electricity, 223
alternating current (AC), 223
characteristics, 223
direct current (DC), 223
motors, 230
characteristics, 231
radial DC motors, 230
solenoids, 231
prototyping, 232 (*See also* Breadboards)
resistors, 224–225
soldering
BlinkM MinM smart LED, 236
learning kits, 238
process, 236–237
SparkFun DS1307 RTC breakouts, 236
tinning, 236
switches
momentary pushbuttons, 229
Reed switches, 230
relays, 230
slide/toggle switches, 230
tilt switches, 230